

DBMS – REQUIREMENT SPECIFICATION

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Skill Task: DBMS

Week: 1

1. Introduction

1.1 Background

EcoCart is an e-commerce platform designed for customers who prefer eco-friendly and sustainable products. The platform provides products that support a greener lifestyle while allowing customers to purchase products through an online ordering system.

The marketplace primarily includes products such as spices and other sustainable products. Customers can browse products, check their prices and availability, add products to a cart, place orders, make payments, and track deliveries.

Managing an e-commerce platform requires maintaining information about customers, products, categories, inventory, orders, payments, shipments, returns, refunds, and reviews. A relational database can be used to store and organize this information and maintain the relationships between the different entities.

1.2 Purpose

The purpose of this document is to specify the business and data requirements of the EcoCart e-commerce order management system. It provides a basis for identifying the entities, attributes, relationships, constraints, and database operations required for the system.

The requirements specified in this document will support the development of:

- Entity-Relationship (ER) model
- Relational schema
- Database tables
- Primary and foreign keys
- Integrity constraints
- Normalized database structure
- SQL queries and CRUD operations

1.3 Problem Statement

An e-commerce system involves several types of related information. Customers may purchase multiple products, and each product may belong to a particular category and have a specific stock quantity. Orders must also be associated with payment, shipment, and delivery information.

If this information is not properly organized, the system may result in:

- Duplicate customer or product information
- Incorrect stock quantities
- Incomplete order records
- Difficulty tracking orders
- Inconsistent payment information
- Difficulty processing returns and refunds
- Difficulty generating business reports

EcoCart addresses these requirements through a structured relational database that maintains customer, product, inventory, order, payment, shipment, return, refund, and review information.

2. Business Requirements Analysis

2.1 Business Objective

The main objective of the system is to provide a structured mechanism for managing transactions between customers and sellers in an online marketplace for handcrafted and artisan products.

The system shall support the following business activities:

1. Customer registration and account management.
2. Seller registration and product listing.
3. Product categorization and management.
4. Product availability and inventory management.
5. Customer shopping cart management.
6. Order creation and order-item management.
7. Payment recording.
8. Shipment and delivery tracking.
9. Return and refund processing.
10. Customer review management.

11. Retrieval of business information through database queries and reports.

2.2 Existing Business Process

In a basic manual process, sellers maintain product information and stock independently. Customers communicate their requirements to the seller and provide order details. Payment and delivery information may be recorded separately.

Such an arrangement creates difficulties when the number of customers and orders increases. Information related to a single transaction may be distributed across multiple records, making it difficult to determine the current status of an order.

2.3 Proposed Business Process

The proposed system provides a common database for storing and managing marketplace information.

The general transaction process is:

Customer → Product Search → Cart → Order → Payment → Shipment → Delivery

If a product needs to be returned:

Delivered Order → Return Request → Return Approval → Refund

The database shall maintain the information generated at each stage.

3. Scope of the System

3.1 In-Scope Functions

The system will include:

Customer Management

- Customer registration
- Customer profile maintenance
- Customer address management
- Customer order history

Seller Management

- Seller registration
- Seller profile maintenance
- Product listing
- Product availability management

Product Management

- Product registration
- Product categorization
- Product price management
- Product description management
- Product availability

Inventory Management

- Stock quantity maintenance
- Stock updates following orders
- Availability checking

Order Management

- Cart management
- Order creation
- Order-item management
- Order status management
- Order history

Payment Management

- Payment recording
- Payment status management
- Payment association with orders

Shipment Management

- Shipment creation
- Shipment status
- Delivery information

Return and Refund Management

- Return request
- Return status

- Refund processing

Review Management

- Product ratings
- Customer reviews

3.2 Out-of-Scope Functions

The following functions are outside the scope of the initial system:

- Actual banking/payment gateway implementation
- Warehouse automation
- Route optimization for delivery vehicles
- Tax calculation based on changing government regulations
- Artificial intelligence-based product recommendations
- Customer chatbot implementation
- External logistics API integration

These functions may be considered as future enhancements.

4. Stakeholders

The major stakeholders of the system are:

<i>Stakeholder</i>	Responsibility
<i>Customer</i>	Searches products, places orders and manages purchases
<i>Seller/Artisan</i>	Lists products and manages product information and stock
<i>Administrator</i>	Manages marketplace information and monitors transactions
<i>Delivery Personnel</i>	Handles shipment and delivery activities
<i>Payment Service</i>	Processes or records payment transactions

The **Customer, Seller, and Administrator** are the primary users of the database system.

5. Functional Requirements

Functional requirements specify what the system shall perform.

FR-01: Customer Registration

The system shall allow a new customer to create an account by providing the required customer information.

The system shall maintain a unique identifier for every customer.

FR-02: Customer Information Management

The system shall allow customer information such as name, contact details, and address to be maintained.

A customer may have more than one delivery address.

FR-03: Seller Registration

The system shall maintain information about sellers or artisans registered with the marketplace.

Each seller shall have a unique seller identifier.

FR-04: Product Management

The system shall allow sellers or authorized administrators to maintain product information including:

- Product name
- Description
- Price
- Category
- Seller
- Availability
- Stock quantity

FR-05: Product Search

The system shall allow customers to identify products based on product name, category, seller, price range, or availability.

FR-06: Category Management

The system shall maintain product categories.

A category may contain multiple products, while each product shall be associated with an appropriate category.

FR-07: Inventory Management

The system shall maintain the available quantity of each product.

Stock quantity shall be updated when products are ordered, cancelled, returned, or restocked.

FR-08: Cart Management

The system shall allow customers to add products to a shopping cart and modify or remove cart items before placing an order.

FR-09: Order Placement

The system shall allow a customer to place an order containing one or more products.

Each order shall be assigned a unique order identifier.

FR-10: Order Item Management

The system shall maintain the individual products included in each order.

For each order item, the system shall maintain information such as:

- Product
- Quantity
- Unit price
- Subtotal

FR-11: Payment Management

The system shall record payment information associated with an order.

The system shall maintain the payment status, transaction identifier, payment amount, and payment date.

FR-12: Shipment Management

The system shall maintain shipment information for orders that are ready for dispatch.

The shipment status shall be updated during the delivery process.

FR-13: Order Tracking

The system shall allow authorized users to determine the current status of an order.

Possible statuses include:

Placed → Confirmed → Shipped → Delivered

An order may also be marked as **Cancelled** where applicable.

FR-14: Return Management

The system shall allow customers to request the return of eligible products.

The system shall maintain the return reason, request date, and return status.

FR-15: Refund Management

The system shall maintain refund information for approved returns.

A refund shall be associated with the relevant return and order.

FR-16: Review Management

The system shall allow customers to provide reviews and ratings for eligible products.

A customer shall be permitted to review a product only when the corresponding purchase has been completed according to the business rules.

6. Non-Functional Requirements

6.1 Data Integrity

The database shall maintain accurate and consistent data.

Primary key and foreign key constraints shall be used to maintain relationships between entities.

6.2 Security

Access to customer, seller, order, and payment information shall be restricted to authorized users.

6.3 Performance

Common operations such as product search, order retrieval, and customer order-history retrieval should be performed within a reasonable response time.

6.4 Reliability

The system should preserve transaction information and prevent loss or duplication of order records.

6.5 Scalability

The database structure should support an increase in customers, sellers, products, and orders without requiring major changes to the database design.

6.6 Maintainability

The database should follow a structured relational design so that modifications and maintenance can be performed without unnecessary duplication of data.

7. Customer CRUD Operations

Create:

The system shall allow authorized users to create customer, seller, product, inventory, order, payment, shipment, return, and refund records.

Read:

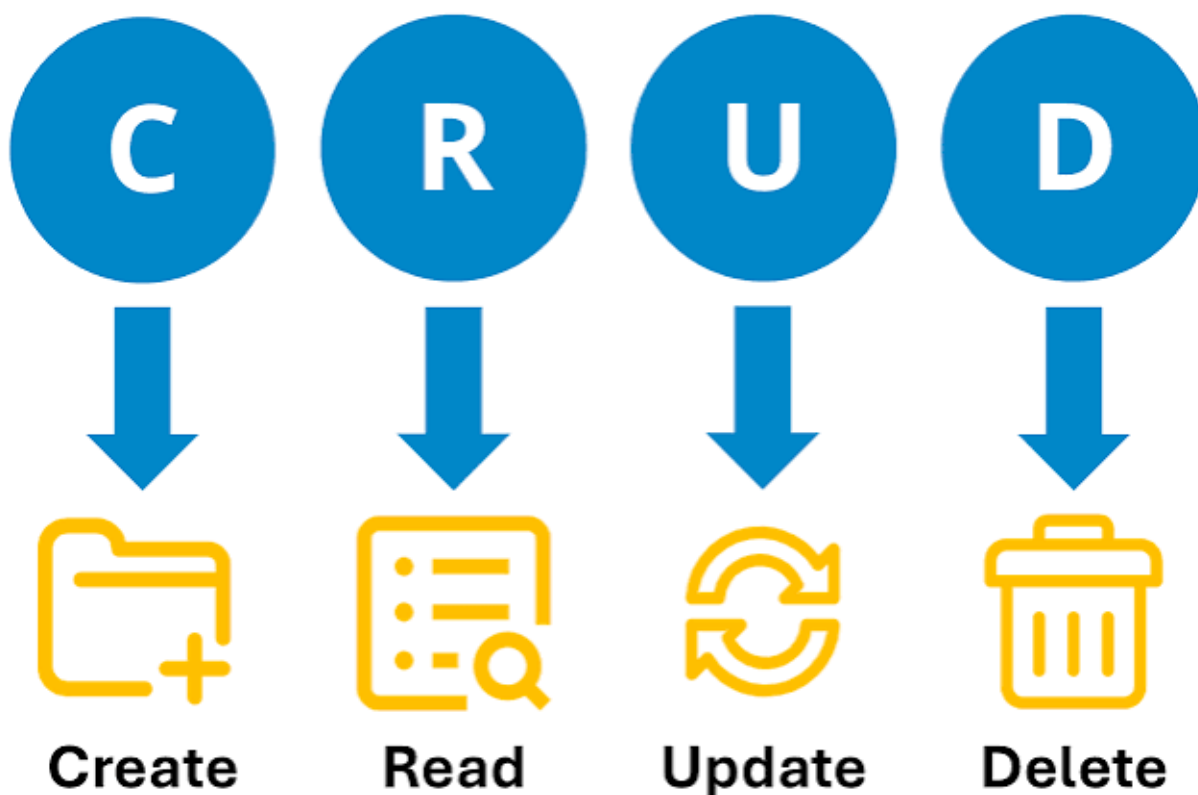
The system shall allow users to retrieve relevant information such as product details, available stock, customer orders, payment status, and shipment status.

Update:

The system shall allow authorized users to update product information, inventory quantities, customer details, order status, payment status, and delivery status.

Delete:

The system shall allow deletion of records where business rules permit. Transactional records such as completed orders and payments should generally be retained rather than physically deleted.



Entity	Create	Read	Update	Delete
Customer	Register customer	View customer details	Update customer profile	Delete customer account

Seller	Register seller	View seller details	Update seller information	Remove seller
Category	Add product category	View categories	Update category details	Delete category
Product	Add sustainable product	View/search products	Update price, description, availability	Remove product
Inventory	Add stock details	Check product stock	Update stock quantity	Remove inventory record
Sustainability Certification	Add certification	View certification details	Update certification information	Remove certification
Cart	Add product to cart	View cart	Update product quantity	Remove product from cart
Order	Place an order	View order/order history	Update order status	Cancel eligible order
Order Item	Add product to order	View items in an order	Update quantity	Remove eligible item
Payment	Record payment	View payment details/status	Update payment status	Remove payment record where permitted
Shipment	Create shipment	Track shipment	Update shipment status	Cancel shipment where permitted
Return	Create return request	View return details/status	Update return status	Cancel eligible return
Refund	Create refund record	View refund details/status	Update refund status	Update refund status / reverse where permitted
Review	Add product review	View reviews	Update review	Delete review

1. CREATE – Table

```
SQL> CONNECT system
Enter password:
Connected.
SQL> CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Email VARCHAR(100) NOT NULL
UNIQUE, Contact_No VARCHAR(15) NOT NULL UNIQUE, Address VARCHAR(200) NOT NULL, Registration_Date DATE NOT NULL);
Table created.
```

2. Add a customer

```
SQL> INSERT INTO Customer (Customer_ID, Customer_Name, Email, Contact_No, Address, Registration_Date) VALUES(105, 'Meena
Devi', 'meena@gmail.com', '9876543214', 'Chennai', '12-MAR-2026');
1 row created.
```

3. READ – Display All Customers

```
SQL> SELECT *
2 FROM Customer;

CUSTOMER_ID
-----
CUSTOMER_NAME
-----
EMAIL
-----
CONTACT_NO
-----
ADDRESS
-----
REGISTRAT
-----
101
```

4. READ – Display a Specific Customer

```
SQL> SELECT *
2 FROM Customer
3 WHERE Customer_ID = 101;
```

5. UPDATE – Update Customer Details

```
SQL> UPDATE Customer
2 SET Contact_No = '9876501234',
3 Address = 'Coimbatore'
4 WHERE Customer_ID = 101;
1 row updated.
SQL> SELECT * from Customer;
```

6. DELETE – Delete a Customer

```
SQL> DELETE FROM Customer
      2 WHERE Customer_ID = 105;

1 row deleted.
```

8. Conclusion

The proposed EcoCart – Sustainable Products E-Commerce Order Management System provides a structured database solution for managing customers, sellers, products, inventory, orders, payments, shipments, returns, refunds, and reviews. The requirement specification identifies the major business processes, data requirements, functional requirements, stakeholders, and CRUD operations of the system.

The requirements defined in this document provide a foundation for designing the Entity-Relationship (ER) model, relational schema, database tables, integrity constraints, and SQL queries. The proposed relational database will help maintain data consistency, reduce duplication, and support efficient retrieval and management of e-commerce transaction information.